Scotty: A Compact System-1 Decision Model Trained with Beam

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Abstract. Many software tasks need a choice rather than a written answer. We introduce Scotty, a compact System-1 decision model that reads a situation, a question and possible answers, then scores the answers in one forward pass. Beam supplies training examples; ModernBERT supplies the pretrained language encoder. Trained on 18,092 decisions from two sources, Scotty achieves 83.03% accuracy on held-out Beam task families and 75.45% on Typed Decisions. A Typed-only control scores 56.34% and 78.35%, respectively. The results show broader coverage across these two collections, with a specialization tradeoff. A local, single-item timing test takes 23.36 ms including text preparation. These single-seed results support compact decision models, with calibration and independent testing still needed.

1 Introduction

A workflow may need to decide whether a parcel is ready to leave, which team should handle a request, or how urgent a maintenance issue is. In each case, the possible answers can be listed in advance. Generating a paragraph and extracting a decision from it adds work that the application may not need.

Scotty is designed for this setting. It reads the available information and selects among supplied options. The same interface supports yes/no questions, choices between named alternatives and ratings on an ordered scale. The application receives scores over those options, rather than free-form text that must be interpreted.

We use *System-1* as an engineering description of this direct decision path: one model pass, without generating a step-by-step explanation. It is not a claim that Scotty reproduces human intuition or that a quick answer is necessarily correct. Hard or unfamiliar cases may still need a larger model or human review.

The central idea is to use a large model's effort when preparing training data, then use a smaller model for repeated decisions. Reflection Beam generates examples with target answers [1]. Scotty learns from these examples alongside the public Typed Decisions dataset [2]. Its 421-million-parameter network starts from ModernBERT [3], with a newly initialized decision head following Laya's design [4]. Beam contributes supervision, not its weights. Our contribution is this implemented decision model and evaluated training recipe, rather than a new Transformer architecture.

2 From a situation to a choice

Every decision has three parts. The *situation* contains the relevant facts. The *question* says what must be decided. The *options* define the allowed answers. Keeping these parts explicit helps an application ask a precise question and obtain a result it can use directly.

Decision type	Example question	Example options
Yes/no	Are the release conditions met?	No; yes
Category	Where should this request go?	Repairs; billing; logistics
Rating	How urgent is this issue?	Low; medium; high; critical

Table 1: Three decision types with one interface. These are illustrative questions, not additional experimental results.

Consider a simple release check:

> **Situation:** A parcel has a shipping label. Its final weight has not been recorded. The release rule requires both a label and a recorded final weight.
>
> **Question:** Is the parcel ready for release?
>
> **Options:** No; yes.
>
> The rule implies "no" because one required condition is missing. This worked example explains the task; it is not a measured Scotty prediction.

Internally, ModernBERT reads the question, options and situation together. It turns the text into numerical representations that reflect the surrounding context. A small decision module then assigns a score to each option. The scores are converted into a probability distribution, and the highest-scoring option gives the discrete answer. No explanation text is generated along this path.

The options are part of the input, not a fixed list built into the last layer. A request can describe the alternatives it wants compared. For ordered ratings, medium instead of high is a one-level error; low instead of critical is three. Flexible options do not guarantee success on new tasks: question clarity, available facts and training coverage still matter.

Scotty has 421,293,827 parameters in total. Its text encoder is the large variant of ModernBERT, and its fresh option-scoring module follows Laya. It does not load trained Beam or Laya weights. Inputs fit within 1,024 tokens, the text pieces processed by the model, with bounded space for questions and options. Long inputs need care because information outside that budget may be lost. "Compact" is relative to the much larger teacher, not a claim of negligible memory use.

3 Learning from Beam and other examples

Beam's role is easiest to understand as preparing practice questions with answers. We ask the teacher to produce operational situations, a decision question, candidate answers and a target choice. The resulting examples cover tasks such as checking prerequisites, routing work and assigning levels of attention. The teacher used for this collection is Reflection Beam 1.0 501B-A23B, accessed through a hosted API [1].

After filtering, the Beam collection contains 19,878 decisions across 18 task families. Checks reject malformed examples, invalid answers and detected duplicates. Similar wording and semantic similarity are also screened, with related examples kept together when creating splits. These checks reduce obvious errors and leakage; they do not establish that every label is correct or every remaining example is semantically unique. The content review was assistant-assisted, not an independent expert annotation study.

The second source, Typed Decisions [2], contains cases from workflows such as customer service, invoices, security and agent traces. One case can contain several questions. Its targets describe a teacher's distribution over the options rather than only one chosen answer. For illustration, a target could put 70% on one answer and 30% on another. Training then preserves that preference instead of treating the second option as entirely unsupported. This is an explanation of the format, not a quoted dataset row.

Beam examples use a single target choice. Both target formats teach the same scoring model to favor the supplied answers. This is transfer through training data, related to knowledge distillation [5]; no Beam layers or token-level probability outputs are copied into Scotty. Generated rationales and gold answers are not included in the model's input.

Source	Training	Validation	Test
Beam-generated	13,292	3,327	3,259
Typed Decisions	4,800	1,200	2,000
Total decisions	18,092	4,527	5,259

Table 2: Training fits the model; validation selects a checkpoint; testing measures the selected model. Test scores remain separate by source.

The Beam test withholds three complete families. They concern operational evidence strength, packing readiness and warehouse exception routing. Training and validation use the other 15 families. For Typed Decisions, all questions from a case stay in the same split. Its official test has 400 cases with 2,000 decisions. These choices make it harder to benefit from near copies of the same situation appearing on both sides of a split, although broader conceptual overlap can remain.

4 How we train and test Scotty

Training combines the 13,292 Beam examples and 4,800 Typed Decisions examples in Table 2. Both the language encoder and decision head are updated to favor the target answers. Three passes through the training set give 849 parameter-update steps. Validation selects the checkpoint after the third pass; this checkpoint stays fixed for every reported evaluation. One random seed is used, so the study does not measure how much another training run might differ. Training preserves the natural source proportions and does not shuffle option order.

4.1 A control that asks a useful question

Does adding Beam examples help, or would training only on Typed Decisions be enough? We train a second model, the *Typed-only control*, with the same starting encoder, fresh head, training settings and random seed. It sees only the 4,800 Typed training decisions. These are repeated until it matches Scotty's 849 updates and 54,276 decision exposures.

This gives both models the same number of update steps and presented decisions, while changing what they learn from. It is not an equal-compute test: examples can have different lengths, and the mixed model sees more unique decisions. Selection gives equal weight to Beam position-balanced accuracy and Typed soft agreement on validation data. Soft agreement credits the teacher probability assigned to the chosen option. Thus Beam validation helps select the control too, although it never trains on Beam examples. The comparison tests the mixed recipe, not only the teacher's effect.

4.2 What the measurements mean

Accuracy is the fraction of decisions where the model's top choice matches the target's top choice. Beam's *position-balanced accuracy* gives equal weight to observed groups defined by decision type, option count and target position: where the correct option appears in the list. This prevents frequent groups from dominating the score. Lower balanced accuracy signals uneven performance across groups.

The two tests ask different questions. The Beam test checks families omitted from training. The Typed test checks new cases within workflows represented during training. Neither is a demonstration of unrestricted real-world generalization. Neither test is used for gradient updates or checkpoint selection, but the tests had been examined in earlier project work; the control comparison is retrospective.

We compare the original released Laya model, called *root Laya*, on the same Beam test. Its training differs and original-data overlap is unknown, so this is a reference point rather than a training control. Uncertainty intervals resample related decisions together by case or generation request. They describe test-sample uncertainty, not variation across training seeds.

5 What the results show

Scotty gets 83.03% of the 3,259 Beam test decisions correct, compared with 56.34% for the Typed-only control and 53.51% for root Laya. The result is encouraging because the three test families were omitted from Scotty's training. Position-balanced accuracy is lower for every model, showing that the raw averages do not tell the whole story.

	Beam family holdout		Typed test
Model	Accuracy	Balanced	Accuracy
Root Laya	53.51	48.56	–
Typed-only control	56.34	51.64	**78.35**
Scotty	**83.03**	**76.39**	75.45

Table 3: Measured accuracy in percent. A dash means no matched result is included here. The stronger model depends on the test collection.

On Typed Decisions, Scotty reaches 75.45%, above the training-derived majority baseline of 48.15%, but below the control's 78.35%. The mixed recipe gains 26.70 percentage points on Beam while giving up 2.90 points on Typed, using unrounded scores. Paired 95% intervals are 24.96–28.42 points for the Beam gain and 1.05–4.80 points for the control's Typed gain. Scotty's accuracy intervals are 81.29–84.75% on Beam and 73.30–77.60% on Typed.

The practical conclusion is a coverage–specialization tradeoff. Beam examples help the model handle this broader operational collection, while repeated Typed-only practice is better for the Typed test. More training sources do not automatically improve every existing task. Published specialized Laya Typed Decisions accuracy is 76.60% [6]; this is context from a different training and evaluation setup, not a matched rerun or evidence that the systems are equivalent.

Ordered ratings remain harder than the other decision types. On Beam, Scotty scores 90.75% for yes/no questions, 91.77% for categories and 62.57% for ordered ratings. On Typed, the corresponding scores are 83.17%, 72.67% and 71.75%. The Typed ordered-rating error averages 0.305 scale positions, with 97.88% of predictions within one level of the target. Being close can be useful, but it is not the same as being correct.

5.1 How fast is one decision

On an NVIDIA GB10, one 350-token, four-option input takes a median 22.21 ms for the model, or 23.36 ms including text preparation. GPU-synchronized timings use batch size one and reduced precision (BF16): five warm-ups, then 20 model-only and ten preparation-inclusive runs. This measures one local decision, not production throughput or a speedup over Beam. Network overhead is excluded; concurrency was not recorded.

6 Practical value and next steps

Scotty turns teacher-generated examples into a reusable decision component. A large model prepares supervision, while a smaller encoder repeatedly scores explicit choices. The results demonstrate one interface for yes/no checks, routing and ordered ratings across two distinct collections.

An application must supply relevant facts, define useful options and check performance on its own cases. High scores should not by themselves authorize actions. Scotty's probabilities are uncalibrated; its raw 15-bin expected calibration error on Typed is 0.1093. Confidence and observed correctness are not reliably interchangeable. Calibration and validation are needed before confidence determines when to act or escalate [7].

Both sources use synthetic supervision, teacher answers can be wrong, and no real-world deployment study is reported. Screening cannot rule out all semantic overlap or pretraining contamination. These single-seed, retrospectively inspected tests do not establish that Beam supplementation is universally best.

Next steps are repeated training, independent tests and compute-matched comparisons, with particular attention to ordered ratings. A handoff to a larger model or human outside Scotty's validated scope remains to be tested. Within these limits, Scotty offers a promising route from Beam-generated supervision to compact, fast decision making.

References

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